



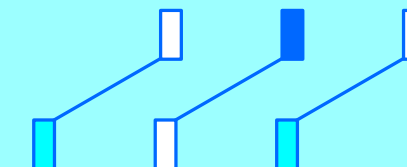
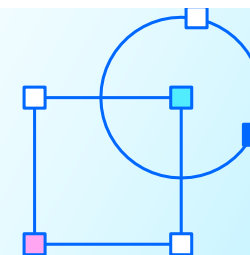
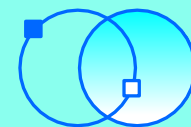
Memento

Private Stories

A focused social space for the people you choose

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Android Developer | Tech Fresher 2026



Agenda

01

Product and foundation

User problem, product scope, architecture and authentication

02

Engineering challenges

Reliable delivery, consistent state and controlled access

03

Solution approach

Data model, media delivery, recovery and access lifecycle

04

Delivery and outcomes

Ownership, delivered outputs and contribution

05

Learning and next steps

Process evolution, roadmap, summary and final demo



SECTION

01

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Product and foundation

The user need, product scope and the shared technical foundation



Some moments need a smaller and more intentional space.

01

FEED DISTRACTION

Personal moments compete with public content

02

AUDIENCE PRESSURE

Broad visibility changes how people share

03

LOST CONTEXT

Shared memories disappear into general timelines

USER NEED**A calm space for chosen connections**

A private social experience built around chosen connections.

01

CHOSEN CONNECTIONS

Invite-based relationships

02

DEFINED AUDIENCE

Every post belongs to one connection

03

FOCUSED EXPERIENCE

No public discovery feed

04

SHARED HISTORY

Moments remain with their context

A private connection space for sharing moments with a chosen audience



One complete journey for private photo sharing

01

IDENTITY

Secure account access and recovery

02

CHOSEN CONNECTION

Invite-based DIRECT sharing

03

PRIVATE STORY

One to five photos with a chosen layout

04

SHARED HISTORY

Realtime updates with offline reading

05

RELIABLE DELIVERY

Background publishing and retry



Delivered foundation

Identity and connection

Authenticated DIRECT sharing

Photo experience

Layouts and on-device processing

Reliable data

Realtime, offline and retry

Access control

Backend validation and rules

Next expansion

Video journey

A separate media and playback contract

Performance evidence

Measure before further optimization

Product expansion

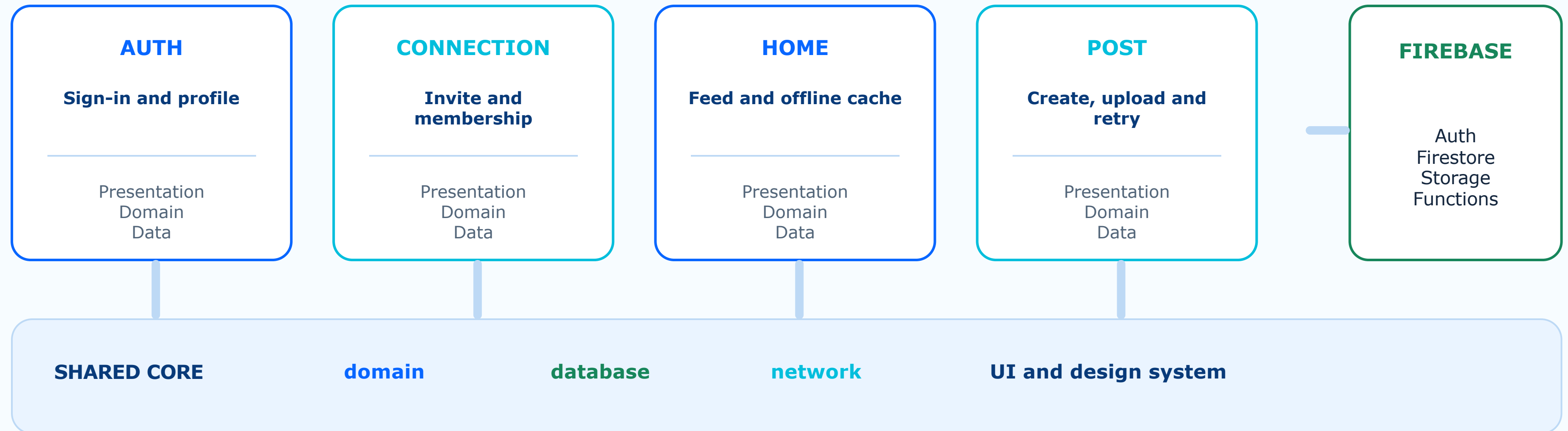
Group, avatar and richer media tools

COMPLETE PHOTO JOURNEY FIRST



APP SHELL

Navigation, Hilt composition and WorkManager scheduling



Why this split

Each feature owns its rules. Core modules provide shared capabilities. Room remains the UI source of truth.



SIGN UP

- 1 Validate input
- 2 Create Firebase Auth account
- 3 Create users/{uid}
- 4 **Cache profile + invite code**

SIGN IN

- 1 Verify email and password
- 2 Restore Firebase session
- 3 Sync profile transaction
- 4 **Cache profile, then open Home**

FORGOT PASSWORD

- 1 Validate email
- 2 Request reset email
- 3 Show a generic confirmation
- 4 **Do not reveal account existence**

Shared identity rule

Firebase Auth UID identifies users/{uid}. Email and password remain in Auth. Splash restores the session; Room is used only for a valid cached profile of the same UID.



SECTION

02

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Engineering challenges

The reliability, consistency and access-control risks behind the core journey



01

Reliable delivery

Continue after UI exit and retry without duplicates

02

Consistent experience

Keep one readable state across online and offline use

03

Controlled access

Apply membership rules throughout the data lifecycle



SECTION

03

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Solution approach

Data ownership, media delivery, recovery and access lifecycle



REMOTE AUTHORITY

Firebase Auth

UID, credentials and session

Firestore

users • connections / members / posts •
private invite data

Storage

JPEG objects scoped by connection, post and
media IDs

Cloud Functions

invite, finalize, delete, disconnect and
cleanup mutations

LOCAL SOURCE OF TRUTH

Room tables

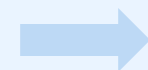
- users
- connections
- connection_members
- posts
- media_items

Private app files

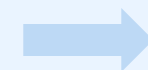
- processed upload drafts
- offline media cache
- Coil memory and disk
cache

**UI reads Room and local media. Remote responses
update local state through repositories.**

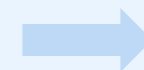
Remote event



Repository



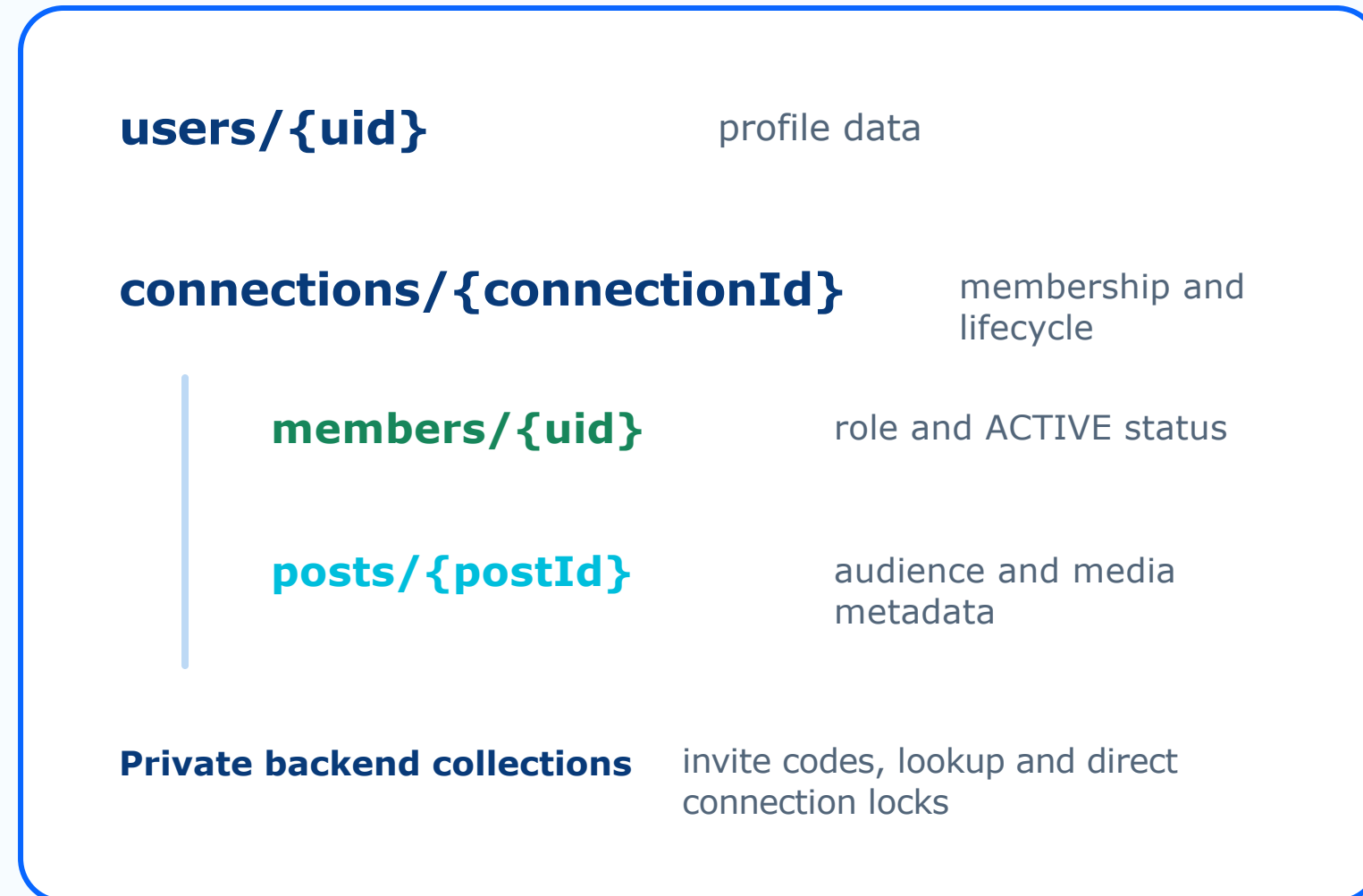
Room transaction



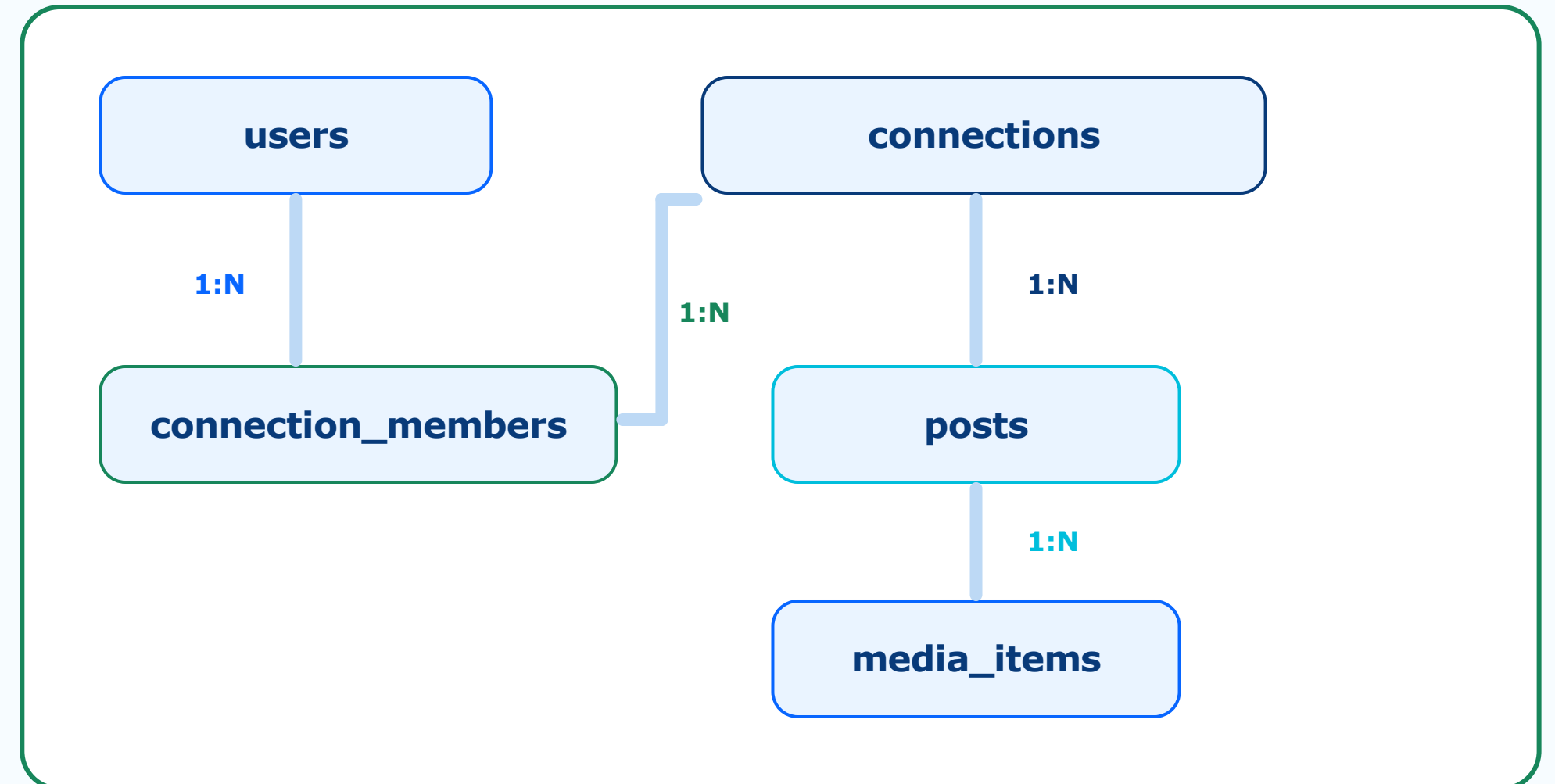
Compose UI



FIRESTORE CONNECTION TREE

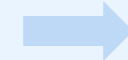


ROOM RELATIONSHIPS

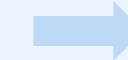


REMOTE MEDIA

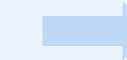
Storage path
scoped by IDs



Offline media cache
≤ 200 MiB



Coil memory 20%
+ disk 100 MiB



UI reads
Room + cache



The schema follows the reads the product performs.

FEED QUERY

memberIds contains uid
status = ACTIVE
createdAt descending
limit + cursor



COMPOSITE INDEX

memberIds	ARRAY_CONTAINS
status	ASCENDING
createdAt	DESCENDING



QUERY RESULT

Audience filtered on the server
Newest posts first
Bounded page size

OTHER ACCESS PATHS

Connection list

memberIds + status + lastPostAt

Deletion sync

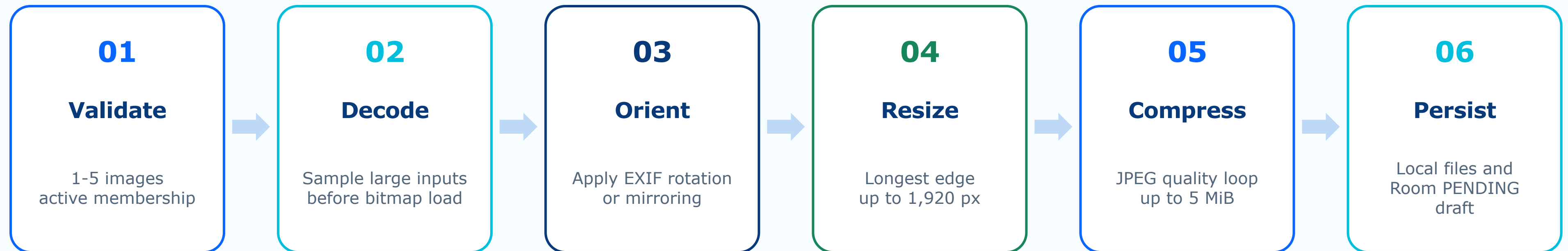
memberIds + status + updatedAt

Connection feed

memberIds + connectionId + status + createdAt

Cost discipline: composite indexes support implemented queries; limits and cursors keep each page bounded.

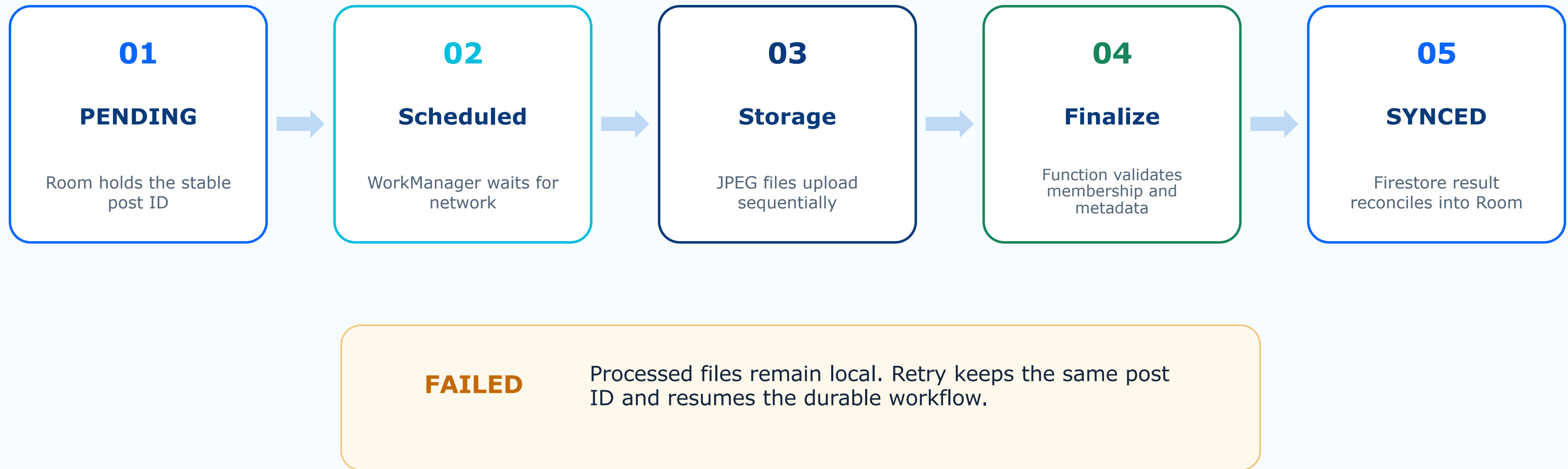


**Why process locally**

Bound memory use, normalize orientation and upload a predictable JPEG payload.

Supported output layouts: **SINGLE • GRID • COLLAGE • CAROUSEL**





Timeouts: up to 2 minutes per file, then 30 seconds for finalize



Failure group	Includes	Current response
Delivery interruption	Network loss or process termination	Persisted draft, network constraint and background retry
Retry and media integrity	Repeated finalize or unavailable media	Stable IDs, backend validation and item-level recovery
Access lifecycle	Delete, disconnect or stale cache	Rules revoke access and tombstones clear app state

Each group becomes a visible state instead of an untracked failure.



01

MEMBERSHIP-SCOPED ACCESS

Only active members can read connection data

02

SERVER AUTHORITY

Audience and sensitive mutations are derived on the backend

03

LIFECYCLE REVOCATION

Delete and disconnect update state before rules deny access

CLIENT INPUT IS NOT AUTHORITY

SECTION

04

/ 05

Delivery and outcomes

How the project was delivered and what the implementation produced



01 Requirements and architecture

Defined project boundaries and module responsibilities.

02 Core user flow

Delivered authentication, connection, post and feed.

03 Reliability hardening

Added durable retry and lifecycle-aware access handling.

04 Final review

Aligned implementation, rules, documentation and demo flow.

Ownership pattern

Define, implement, verify and document, then close the loop



Android product

The complete private photo-sharing journey

Cloud services

Server-authoritative mutations and media lifecycle

Offline data layer

Local state, realtime reconciliation and retry

Operational handoff

Access rules, configuration and project documentation

One product flow delivered across client, cloud and local state



IMPACT

Impact and contribution

SITUATION

Upload and access lifecycle had multiple failure paths across UI, local state and Firebase.

BEHAVIOR

Implemented durable drafts, background retry, server validation and lifecycle-aware access.

IMPACT

The core flow now builds, retries without duplicate identity and revokes remote access after lifecycle changes.

Outcome: the core journey remains reliable across normal and interrupted states



SECTION

05

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Learning and next steps

Applied feedback, current boundaries and the order of future development



01

Scope first

From a feature list to one complete photo journey

02

State ownership

From screen-owned work to durable local state

03

Failure-first design

From happy path to retry, recovery and revocation

04

Delivery discipline

Align implementation, rules, documentation and demo



Measure the foundation before expanding the product.

01

Release readiness

Signed build and CI
Crash visibility
Performance baseline

02

Experience efficiency

Viewport prefetch
Measured cache policy
Device profiling

03

Product expansion

Complete video journey
Media enhancements
Connection features

One complete journey at a time



01

Private by design

Sharing stays inside chosen connections.

02

Reliable experience

Drafts, retry and offline access preserve continuity.

03

Ready to evolve

Modular boundaries and query-driven data support the next journey.



LIVE DEMO

Connect → Layouts → Background publish → Realtime/offline → Retry

1

Connect A and B

2

Switch photo
layouts

3

Publish in
background

4

Realtime + offline
view

5

Recover a failed
draft

THANK YOU

Q&A

Tang Hong Ai | Memento

